

E - Team Division

分值: 475 分

Problem Statement

Divide N players, player 1, player 2, ..., player N , into two (distinguishable) teams A and B satisfying all of the following conditions.

将 N 名选手 (选手 1、选手 2、...、选手 N) 分成两个可区分的队伍 A 和 B , 满足以下所有条件。

- Each team consists of at least one player.
 - Each player belongs to either team A or team B , but not both.
 - The number of players in the team that player i belongs to is at least L_i and at most R_i .
 - 每支队伍至少包含一名选手。
 - 每名选手属于队伍 A 或队伍 B 中的恰好一个, 不能同时属于两个队伍。
 - 选手 i 所在的队伍的选手人数至少为 L_i , 至多为 R_i 。
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Find the number of ways to divide the players satisfying the conditions, and output the count modulo 998244353.

Two divisions are considered different if there exists a player who belongs to different teams in the two divisions.

求满足条件的选手分队方案数, 结果对 998244353 取模后输出。如果存在至少一名选手在两种分队方案中属于不同的队伍, 则认为这两种分队方案是不同的。

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq L_i \leq R_i \leq N - 1$
- All input values are integers.
- $2 \leq N \leq 2 \times 10^5$
- $1 \leq L_i \leq R_i \leq N - 1$

- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N
 $L_1 R_1$
 $L_2 R_2$
 $\vdots$
 $L_N R_N$

Output

Output the number of valid divisions, modulo 998244353.

输出合法的分队方案数，对 998244353 取模。

Sample Input 1

```
3
1 1
1 2
2 2
```

Sample Output 1

```
2
```

The following two divisions satisfy the conditions.

以下两种分队方案满足条件。

- Player 1: Team A , Player 2: Team B , Player 3: Team B
- Player 1: Team B , Player 2: Team A , Player 3: Team A

- 选手 1: 队伍 A , 选手 2: 队伍 B , 选手 3: 队伍 B
- 选手 1: 队伍 B , 选手 2: 队伍 A , 选手 3: 队伍 A

Since 2 modulo 998244353 is 2, output 2.

由于 2 模 998244353 的结果是 2, 输出 2。

Sample Input 2

```
6
1 5
1 5
2 5
1 3
3 5
2 5
```

Sample Output 2

```
30
```

There are 30 valid divisions.

共有 30 种合法的分队方案。